<u>JPR Assignment 5</u>

1. List constructors of ServerSocket class.

Ans.

1. ServerSocket() throws IOException
2. ServerSocket(int port) throws IOException
3. ServerSocket(int port, int backlog) throws IOException
4. ServerSocket(int port, int backlog, InetAddress bindAddr) throws IOException

2. Define ProxyServer and Reverse socket.

Ans.

1) Proxy Server: A proxy server acts as an intermediary between a client and the internet or another server. It forwards client requests to the target server and then returns the server's response to the client. Proxy servers help improve security, control access, and cache data to speed up requests.

2) Reverse Socket: A reverse socket (often called a reverse shell) is a connection where a remote device connects back to a local machine to provide access or control. It is commonly used for remote administration or troubleshooting when the local machine is behind a firewall or NAT, allowing the remote system to initiate the connection.

3. Differentiate between TCP and UDP.

Ans.

TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
TCP is **connection-oriented.**	UDP is **connectionless.**
It is **reliable** as it ensures data delivery.	It is **unreliable**, delivery is not guaranteed.
Data is sent in **sequence.**	Data may arrive **out of order.**
Slower due to error checking and correction.	**Faster** as there is no error checking.
Used for applications like **web browsing, email.**	Used for **video streaming, gaming.**
Header size is **larger** (20–60 bytes).	Header size is **smaller** (8 bytes).

4. WAP using URL class to retrieve the host. protocol, port and file of URL http://www.msbte.org.in

Ans.

```
import java.net.URL;
public class URLInfo
{
public static void main(String[] args)
{
try
{
URL url = new URL("http://www.msbte.org.in");
System.out.println("Host: " + url.getHost());
System.out.println("Protocol: " + url.getProtocol());
System.out.println("Port: " + url.getPort());
System.out.println("File: " + url.getFile());
}
catch (Exception e)
{
e.printStackTrace();
}
}
}
```